Akshath Raghav Ravikiran

765-404-8121 | araviki@purdue.edu | linkedin.com/in/akshathrr | akshathraghav.github.io

Summary

MS-ECE candidate targeting early-career roles in RTL design, (micro)architecture modeling, and HW/SW co-design for AI ASICs and ML systems.

Education

Purdue University, West Lafayette — **Cumulative GPA: 3.64**

Master of Science in Electrical and Computer Engineering — *January 2026 – August 2026*

Bachelor of Science in Computer Engineering — *August 2022 – December 2025*

Coursework: Accelerator Architectures, Computer Architecture, MOS VLSI Design, AI Hardware Design, ASIC Design, Operating Systems, Networks

Honors: ECE Senior Design Award, Eli Shay Scholarship, UIUC HDR Fellowship, Outstanding Sophomore (VIP), Dean's List & Semester Honors (7x)

Organizations: IEEE Eta Kappa Nu, Purdue SoCET, ECE Graduate Student Association, Purdue Chess Club

Technical Skills

RTL Design & Verification – SystemVerilog, UVM | Simulation (QuestaSim, Verilator) | FPGA (Vivado, ProtoCompiler) | ASIC Synthesis (Genus)
Architecture & Modeling – C/C++, RISC-V Assembly | Simulators (gem5, GPGPU-Sim, Ramulator2) | Profiling (AMD uProf)
Compute & Embedded – Python, Bash, TCL Scripting | GPU Programming (CUDA, Triton) | AI/ML (PyTorch, TensorFlow) | ESP-IDF

Experience

Graduate Research Assistant — **January 2026 – Present**

Purdue SoCET (PI: Prof. Mark Johnson, Purdue-ECE) — *West Lafayette, IN, US*

- AI Hardware: Architecting HW/SW stack for **system-level simulation** of a custom DNN Accelerator – Ax01 "Atalla" Tensor Core.
  - Leading development of a non-blocking AXI4 bus with a custom DDR4 controller; integrating HBM3 behavior w/ Ramulator2.
  - Helping develop custom PyTorch backend that lowers ML models into compiled micro-kernels via a custom ISA and compiler.
- GPU: Directing RTL Design of Cx01 "Cardinal" GPU Core for graphics workloads. Designing on-chip caches (+ T-Cache, SMEM).
- AI Applications: Technical Lead for developing a LangGraph-based agentic-platform for internal use in functional RTL verification.
- Datacenter Acceleration: Supporting verification of a custom RoCE v2 NIC by automating emulation flows on the HAPS FPGA.

Undergraduate Research Assistant — **July 2024 – December 2025**

Purdue SoCET (PI: Prof. Mark Johnson, Purdue-ECE) — *West Lafayette, IN, US*

- AI Hardware: Leading on-chip Memory Subsystem for Ax01 "Atalla" Tensor Core; focusing on architecture diagramming & ISA design.
  - Built a cycle-accurate simulator of the datapath for **performance modelling** using implicit-convolution and GEMM kernels.
  - Architected a parameterizable Scratchpad with on-the-fly swizzling and a pipelined $N \times N$ crossbar — optimized for PPA.
- GPU: Implemented a lockup-free-cache cache – Achieved 100% coverage (ModelSim); optimized to synthesize (Genus) at 700MHz.
- Enhanced the AFTx07 RISC-V core with Zicond extension for macro-fusion of conditional arithmetic/logic sequences.

Undergraduate ML Researcher — **August 2023 – April 2024**

Duality Lab (Contract w/ Google LLC – PI: Prof. James Davis, Purdue) — *West Lafayette, IN, US*

- Helped re-engineer the MaskFormer model from PyTorch to TensorFlow to run on GCP TPUs & integrate into the TF Model Garden.
- Contributed to a technical white paper, providing implementation guidance for TPU-focused **HW/SW co-design**.
- Integrated auxiliary losses & MLOps-based tuning on GCP to increase Panoptic Quality scores by 21% (avg.) on the COCO Dataset.

AI Engineering Intern — **May 2025 – August 2025**

BMW - Group IT — *Munich, Bavaria, Germany*

- Built remote AI Agents w/ LangGraph to crawl enterprise apps & automatically run QA tests – eliminating Playwright automation KPIs.
- Deployed a GitHub Copilot Assistant for human-in-the-loop Azure infra provisioning, enabling front-end developers to deploy apps.
- Deployed FastMCP connectors for core DevOps platforms into GAIA (internal AI platform) w/ AWS Lambda & EventBridge.
- Engineered **RAG connectors** exposing knowledge graphs into IDEs – outperformed GAIA (retrieval accuracy) on 10K+ LOC repos.

Selected Projects

BoilerNet – Compute-Enabled Mini-NAS — *KiCad 9.0, PCB Bring-Up, Edge ML*

- Designed PCBs, assembled into 3D-printed enclosures, to form a Network Attached Storage w/ swappable compute blades & memory slots.
- Received Purdue ECE's Senior Design Award in Spring '25 for our **decentralized MCU** stack & quantized-MobileNetV2 inference pipelines.

RISC-V Five-Stage Multicore Processor — *QuestaSim, Xilinx Vivado, FPGA Prototyping*

- Developed a pipelined processor implementing branch prediction, hazard detection logic, and dual-core MSI snoopy cache coherency.
- Synthesized to DE2-115 FPGA at 60MHz — Performed **static timing analysis** showing 2.77× speedup over single-cycle design.

gem5 Microarchitecture Studies — *gem5, Architectural Simulation*

- Added a Waiting Instruction Buffer into the **O3 CPU** to offload load-dependents, improving IPC by up to 13% on SPEC benchmarks.
- Implemented a victim cache with no-allocate and mostly-exclusive policies, reducing L1 miss rates by up to 5% on high-locality workloads.

8-bit Wallace Tree Multiplier – Physical Design — *Cadence Virtuoso ADE, GPDK45nm*

- Designed a Wallace Tree Multiplier using the full-adder inversion property, saving 298 transistors and achieving a 3305 μm^2 footprint.
- Completed **schematic-to-layout flow** with DRC/LVS and post-layout parasitic extraction, with 1.74 ns delay and 320 fJ energy (1.0 V).

tinySpeech – Speech Recognition on Edge Devices — *PyTorch, ML Quantization, MCU Programming*

- Reproduced TinySpeech word-recognition models; Achieved 91% precision benchmarks w/o access to DarwinAI's proprietary code.
- Developed **quantization-aware training** pipelines allowing for INT4/8 training and embedded-C inference engine for RISC-V architecture.